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# Remote Development Project Report

**Student Name:** Qi Yongpeng  
**Student ID:** ZY2557206

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## System Configuration

The development and testing of this project were conducted on a Linux-based environment. The system configuration is summarized in the table below.

| Item                     | Configuration               |
|--------------------------|-----------------------------|
| CPU Model                | Intel(R) Core(TM) i7-11800H |
| Memory Size              | 16GB                        |
| Operating System Version | Ubuntu 22.04 LTS            |
| Compiler Version         | GCC 11.4.0                  |
| Python Version           | Python 3.10                 |

The following command line tools were used to obtain system information:

```
lscpu
free -h
uname -a
gcc --version
python --version
```

These commands allow users to inspect hardware specifications and software environment details directly from the Linux terminal.

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# Implementation Details

Matrix multiplication is an important operation in linear algebra and is widely used in scientific computing, machine learning, and data analysis.

Given two matrices:

- Matrix **A** with dimension  $m \times n$
- Matrix **B** with dimension  $n \times p$

The resulting matrix **C** will have dimension  $m \times p$ .

Each element of matrix **C** is calculated as:

$$C[i][j] = \sum(A[i][k] * B[k][j])$$

This formula means that the element in row  $i$  and column  $j$  of the result matrix is obtained by multiplying corresponding elements of row  $i$  in matrix A and column  $j$  in matrix B, then summing the results.

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## Python Language Implementation

### Source Code

The matrix multiplication algorithm was implemented using Python.  
The program performs multiplication using three nested loops.

```

# matrix_multiplication.py

def matrix_multiply(A, B):

    rows_A = len(A)
    cols_A = len(A[0])
    cols_B = len(B[0])

    # initialize result matrix
    result = [[0 for _ in range(cols_B)] for _ in range(rows_A)]

    # matrix multiplication
    for i in range(rows_A):
        for j in range(cols_B):
            for k in range(cols_A):
                result[i][j] += A[i][k] * B[k][j]

    return result

# example matrices (3x3 example)

A = [
    [2, 1, 3],
    [0, 4, 5],
    [7, 2, 6]
]

B = [
    [1, 3, 2],
    [4, 0, 1],
    [5, 2, 3]
]

C = matrix_multiply(A, B)

print("Result Matrix:")
for row in C:
    print(row)

```

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## Execution Command

The program can be executed using the following command in the terminal:

```
python matrix_multiplication.py
```

Expected output:

```
Result Matrix:  
[21, 12, 14]  
[41, 10, 19]  
[45, 33, 34]
```

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## Algorithm Verification

To verify the correctness of the implementation, a larger  $3 \times 3$  matrix example was used.

Matrix A:

```
2 1 3  
0 4 5  
7 2 6
```

Matrix B:

```
1 3 2  
4 0 1  
5 2 3
```

Manual calculation for the first row:

```
C[1][1] = 2×1 + 1×4 + 3×5 = 21  
C[1][2] = 2×3 + 1×0 + 3×2 = 12  
C[1][3] = 2×2 + 1×1 + 3×3 = 14
```

Second row:

```
C[2][1] = 0×1 + 4×4 + 5×5 = 41  
C[2][2] = 0×3 + 4×0 + 5×2 = 10  
C[2][3] = 0×2 + 4×1 + 5×3 = 19
```

Third row:

```
C[3][1] = 7×1 + 2×4 + 6×5 = 45  
C[3][2] = 7×3 + 2×0 + 6×2 = 33  
C[3][3] = 7×2 + 2×1 + 6×3 = 34
```

The calculated results match the output generated by the Python program, confirming the correctness of the matrix multiplication algorithm.

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## Conclusion

In this project, several important concepts in software development and system operation were explored.

First, the Unix/Linux command line was used to inspect system configurations and execute programs efficiently.

Second, Markdown was used to create structured technical documentation that can be easily converted into other formats such as PDF.

Third, a matrix multiplication algorithm was implemented using Python and verified with test cases.

Through this assignment, I gained practical experience with command line tools, documentation practices, and algorithm implementation using an interpreted programming language.

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## References

1. Python Official Documentation  
<https://docs.python.org/>
  2. Markdown Guide  
<https://www.markdownguide.org/>
  3. Linux Command Line Documentation  
<https://linuxcommand.org/>
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## Appendix

### Additional Notes

Matrix multiplication is a computationally intensive operation for large matrices.

Future work could involve implementing the algorithm using a compiled language such as C or C++ and comparing the execution performance between interpreted and compiled languages.